

Wrong Centre, Wrong Vector, Wrong Scale Factor

Find the mistake, fix the number, and name what was misunderstood

- Translation vector from P to P' : each component is the image coordinate *minus* the original.
 - Dilation about the origin: $(x, y) \rightarrow (kx, ky)$. About a centre (h, j) : $(x, y) \rightarrow (h + k(x - h), j + k(y - j))$.
 - Scale factor = $\frac{\text{image length}}{\text{original length}}$, and areas of similar figures scale by k^2 .
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1. A translation maps $A(3, -2)$ to $A'(7, 1)$. Ana writes the rule as $(x, y) \rightarrow (x - 4, y - 3)$.

(a) Find the correct horizontal component of the translation. _____

(b) Find the correct vertical component. _____

(c) Explain what Ana did wrong, and say what her rule would do to A .

2. A dilation with centre the origin and scale factor 3 should map $B(-2, 5)$ to B' . Ben reports $B'(1, 8)$, because he added 3 to each coordinate.

(a) Write the correct coordinates of B' . $B' (\rule{1cm}{0.4pt}, \rule{1cm}{0.4pt})$

(b) Explain what Ben did wrong, and name the transformation he actually carried out.

3. Two similar triangles have corresponding sides of 9 cm and 15 cm. Cara says the scale factor from the smaller triangle to the larger one is 6, because $15 - 9 = 6$.

(a) Find the correct scale factor, to the nearest hundredth. $\rule{1cm}{0.4pt}$

(b) Explain why a difference of two side lengths can never be a scale factor.

4. A dilation with centre $C(1, 4)$ and scale factor 2 maps $P(5, 6)$ to P' . A student ignored the given centre, used the origin instead, and reported $P'(10, 12)$.

(a) Write the correct coordinates of P' . $P' (\rule{1cm}{0.4pt}, \rule{1cm}{0.4pt})$

(b) Explain what the centre of a dilation controls, and what goes wrong when the origin is used in its place.

5. Two similar rectangles have areas 32 cm^2 and 72 cm^2 . A student says the linear scale factor from the smaller to the larger is 2.25, because $72 \div 32 = 2.25$.

(a) Find the correct linear scale factor. _____

(b) Explain why the ratio of the areas is not the linear scale factor.

6. Segment $\overline{MN}$ has endpoints $M(2, 3)$ and $N(6, 3)$. It is dilated by a scale factor of 3 about the point M , so M itself does not move. A student dilated about the origin instead and reported $M'(6, 9)$ and $N'(18, 9)$.

(a) A dilation about M leaves the y -coordinate at 3. Find the correct x -coordinate of N' .

(b) Find the length of the correct image segment $\overline{M'N'}$. _____

(c) The student's image segment also has the correct length. Explain why the length came out right while the position did not.